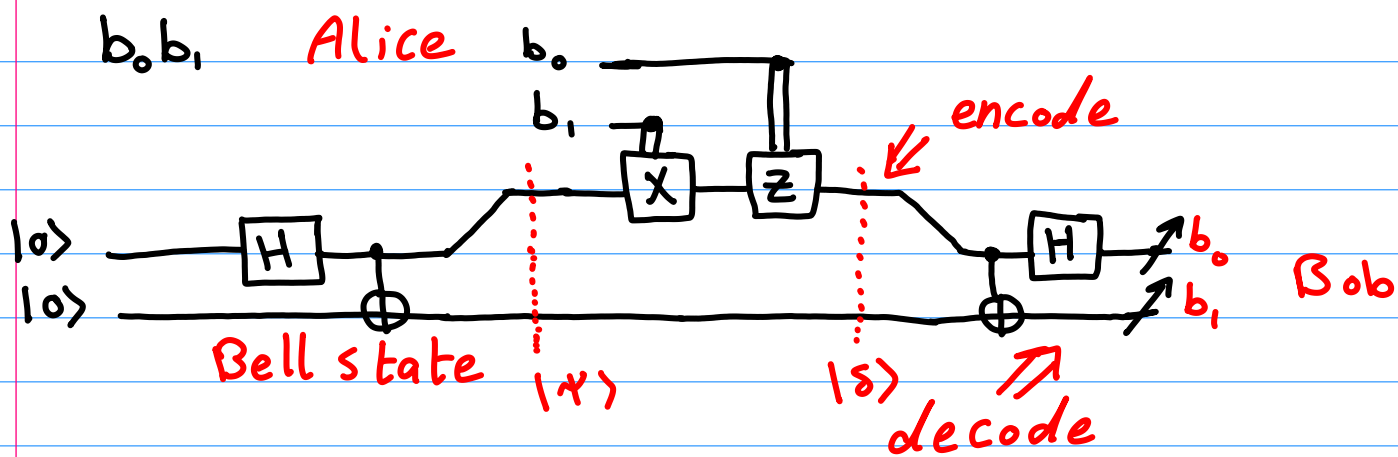


Super dense coding:

Transmit 2 bits at the same time
via encoding them into one quantum
state then decode it.



$$|\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$b_0 b_1 = 00$$

$$|\psi'\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad \text{CNOT}$$

$$= \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

$$= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |0\rangle \xrightarrow{H} |+\rangle |0\rangle = |00\rangle$$

$b_0 b_1$

$$b_0 b_1 = 01$$

$$|s\rangle = \frac{1}{\sqrt{2}} (|10\rangle + |01\rangle) \quad \text{CNOT}$$

$$= \frac{1}{\sqrt{2}} (|11\rangle + |01\rangle)$$

$$= \frac{1}{\sqrt{2}} (|1\rangle + |0\rangle) |1\rangle$$

$$= |+\rangle |1\rangle$$

H ↗

$$= |01\rangle$$

$b_0 b_1$

$$b_0 b_1 = 10$$

$$|s\rangle = \frac{1}{\sqrt{2}} (|100\rangle - |11\rangle) \quad \text{CNOT}$$

$$= \frac{1}{\sqrt{2}} (|100\rangle - |110\rangle)$$

$$= \frac{1}{\sqrt{2}} (|10\rangle - |11\rangle) |0\rangle$$

H ↘

$$= |-\rangle |0\rangle$$

$$= |10\rangle$$

$b_0 b_1$

$$b_0 b_1 = 11$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad X$$

$$= \frac{1}{\sqrt{2}} (|10\rangle + |01\rangle) \quad Z$$

$$|\phi\rangle = \frac{1}{\sqrt{2}} (-|10\rangle + |01\rangle) \quad CNOT$$

$$= \frac{1}{\sqrt{2}} (-|11\rangle + |01\rangle)$$

$$= \frac{1}{\sqrt{2}} (-|1\rangle + |0\rangle) |1\rangle$$

$$H \Rightarrow = |-\rangle |1\rangle$$

$$= \underbrace{|11\rangle}_{b_0 b_1}$$

so this protocol doubles the speed of transmitting bits.

$$\begin{array}{c} a \\ b \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \oplus \end{array} \begin{array}{c} a \\ b \oplus a \end{array}$$

$$b=0$$

$$\begin{array}{c} a \\ 0 \oplus a = a \end{array}$$

$$\text{so } \begin{array}{c} a \\ 0 \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \oplus \end{array} \begin{array}{c} a \\ a \end{array}$$

$$b=1$$

$$\begin{array}{c} a \\ 1 \oplus a = \bar{a} \end{array}$$

$$\text{so } \begin{array}{c} a \\ 1 \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \oplus \end{array} \begin{array}{c} a \\ \bar{a} \end{array}$$

$$\begin{array}{c} a \\ b \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \oplus \end{array} \begin{array}{c} a \\ (-1)^{ab} b \end{array}$$

add -1 phase
if $ab=1$

$$|a\rangle|b\rangle \Rightarrow (-1)^{a \cdot b} |a\rangle|b\rangle$$

$$H|\alpha\rangle = \frac{|0\rangle + (-1)^\alpha |1\rangle}{\sqrt{2}}$$

$$\alpha = \{0, 1\}$$

$$\text{for } \alpha=0$$

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$\text{for } \alpha=1$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Advanced proof:

b_0, b_1 unknown

$$|\gamma\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad \text{for } b_0 = 0 \quad I$$

$$\begin{aligned} &= \frac{1}{\sqrt{2}} (|0 \oplus b_0\rangle |0\rangle + |1 \oplus b_0\rangle |1\rangle) \quad b_0 = 1 \quad X \\ &= \frac{1}{\sqrt{2}} (|b_0\rangle |0\rangle + |\bar{b}_0\rangle |1\rangle) \quad \text{like CNOT} \end{aligned}$$

$\begin{matrix} a \\ b \end{matrix} \begin{matrix} \text{---} \\ \oplus \end{matrix} \begin{matrix} a \\ b \oplus a \end{matrix}$

$$|\delta\rangle = \frac{1}{\sqrt{2}} \left((-1)^{b_1 b_0} |b_0\rangle |0\rangle + (-1)^{b_1 \bar{b}_0} |\bar{b}_0\rangle |1\rangle \right) \quad \begin{matrix} b_1 = 0 & I \\ b_1 = 1 & Z \end{matrix}$$

Note: $(-1)^{\bar{b}_0} = -(-1)^{b_0}$

$$|\delta\rangle = \frac{1}{\sqrt{2}} \left((-1)^{b_1 b_0} |b_0\rangle |0\rangle + (-1)^{b_1} (-1)^{b_1 b_0} |\bar{b}_0\rangle |1\rangle \right)$$

$$= \frac{1}{\sqrt{2}} * (-1)^{b_1 b_0} (|b_0\rangle |0\rangle + (-1)^{b_1} |\bar{b}_0\rangle |1\rangle)$$

Apply CNOT

$$\frac{1}{\sqrt{2}} * (-1)^{b_1 b_0} (|b_0\rangle |0 \oplus b_0\rangle + (-1)^{b_1} |\bar{b}_0\rangle |1 \oplus \bar{b}_0\rangle)$$

$$= \frac{1}{\sqrt{2}} * (-1)^{b_1 b_0} (|b_0\rangle |b_0\rangle + (-1)^{b_1} |\bar{b}_0\rangle |b_0\rangle)$$

$$= \frac{1}{\sqrt{2}} * (-1)^{b_1 b_0} (|b_0\rangle + (-1)^{b_1} |\bar{b}_0\rangle) |b_0\rangle$$

Apply H on q_0

$$H |b_0\rangle = \frac{|0\rangle + (-1)^{b_0} |1\rangle}{\sqrt{2}}$$

$$H |\bar{b}_0\rangle = \frac{|0\rangle + (-1)^{\bar{b}_0} |1\rangle}{\sqrt{2}} = \frac{|0\rangle - (-1)^{b_0} |1\rangle}{\sqrt{2}}$$

$$H (|b_0\rangle + (-1)^{b_1} |\bar{b}_0\rangle) =$$

$$\frac{1}{\sqrt{2}} (|0\rangle + (-1)^{b_0} |1\rangle + (-1)^{b_1} |0\rangle - (-1)^{b_1 b_0} |1\rangle)$$

$$= \frac{1}{\sqrt{2}} [(1 + (-1)^{b_1}) |0\rangle + (-1)^{b_0} (1 - (-1)^{b_1}) |1\rangle]$$

$$= \frac{1}{\sqrt{2}} [2(1 - b_1) |0\rangle + (-1)^{b_0} * 2 b_1 |1\rangle]$$

$$\begin{aligned}
|\alpha\rangle &= \frac{1}{2} * (-1)^{b_1 b_0} \left[\underline{2} (1-b_1) |0\rangle + (-1)^{b_0} * \underline{2} b_1 |1\rangle \right] |b_0\rangle \\
&= (-1)^{b_1 b_0} \left[(1-b_1) |0\rangle + (-1)^{b_0} b_1 |1\rangle \right] |b_0\rangle \\
&= \left[(-1)^{b_1 b_0} \underline{(1-b_1)} |0\rangle + (-1)^{b_0(1+b_1)} \underline{b_1} |1\rangle \right] |b_0\rangle
\end{aligned}$$

b_1 is selector $b = \{0, 1\}$

$$\text{if } b_1 = 0 \quad (-1)^0 (1-0) |0\rangle + 0 = |0\rangle$$

$$\text{if } b_1 = 1 \quad 0 + (-1)^{2b_0} * 1 * |1\rangle = |1\rangle$$

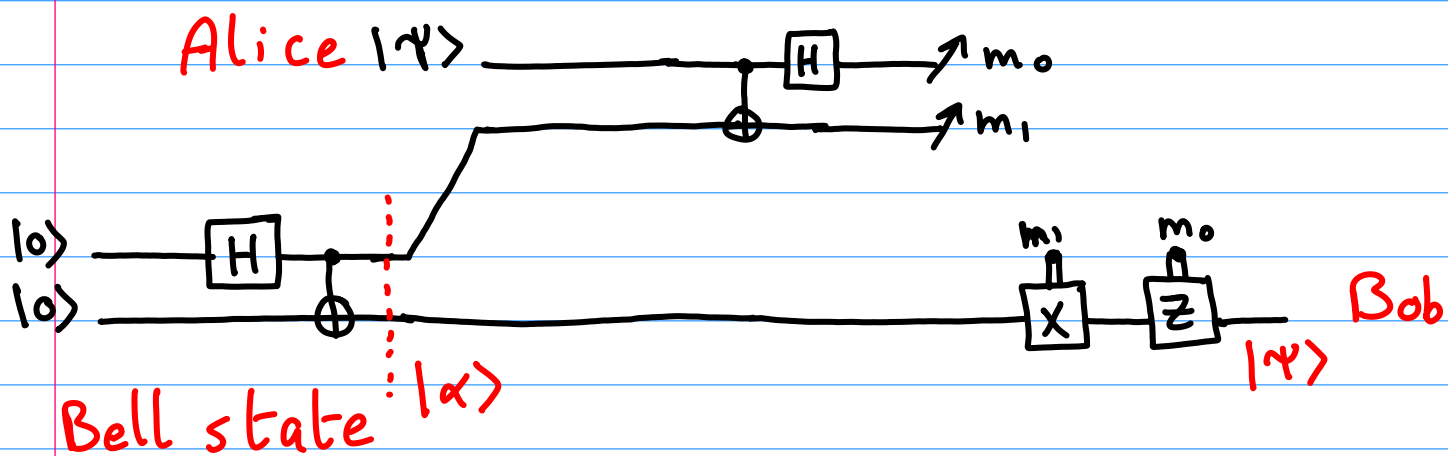
so the whole term is $|b_1\rangle$

$$|\alpha\rangle = |b_1 b_0\rangle$$

Quantum Teleportation:

Teleport one unknown qubit state from Alice and Bob without sending the state itself in a direct connection.

it uses entanglement and classical connection.



Alice in black

Bob in red

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

$$|\psi\rangle|\alpha\rangle = (a|0\rangle + b|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle|0\rangle + |1\rangle|1\rangle)$$

$$\frac{1}{\sqrt{2}} (a|00\rangle|0\rangle + a|01\rangle|1\rangle + b|10\rangle|0\rangle + b|11\rangle|1\rangle)$$

Apply CNOT

$$\frac{1}{\sqrt{2}} (a|00\rangle|0\rangle + a|01\rangle|1\rangle + b|11\rangle|0\rangle + b|10\rangle|1\rangle)$$

Apply H

$$= \frac{1}{2} \left[(a|0\rangle + a|1\rangle)|0\rangle|0\rangle + (a|0\rangle + a|1\rangle)|1\rangle|1\rangle + \right. \\ \left. (b|0\rangle - b|1\rangle)|1\rangle|0\rangle + (b|0\rangle - b|1\rangle)|0\rangle|1\rangle \right]$$

$$= \frac{1}{2} \left[a|00\rangle|0\rangle + a|10\rangle|0\rangle + a|01\rangle|1\rangle + a|11\rangle|1\rangle + \right. \\ \left. b|01\rangle|0\rangle - b|11\rangle|0\rangle + b|00\rangle|1\rangle - b|10\rangle|1\rangle \right]$$

$$= \frac{1}{2} \left[|00\rangle(a|0\rangle + b|1\rangle) + |01\rangle(a|1\rangle + b|0\rangle) + \right. \\ \left. |10\rangle(a|0\rangle - b|1\rangle) + |11\rangle(a|1\rangle - b|0\rangle) \right]$$

Apply Measurement

Collapse into 4 options m_0, m_1
(00, 01, 10, 11)

Applying correction

For $m_0 m_1 = 00$

$$a|0\rangle + b|1\rangle = |\psi\rangle$$

For $m_0 m_1 = 01$

Apply X

$$a|1\rangle + b|0\rangle \Rightarrow a|0\rangle + b|1\rangle = |\psi\rangle$$

For $m_0 m_1 = 10$

Apply Z

$$a|0\rangle - b|1\rangle \Rightarrow a|0\rangle + b|1\rangle = |\psi\rangle$$

For $m_0 m_1 = 11$

Apply X then Z

$$a|1\rangle - b|0\rangle \Rightarrow a|0\rangle - b|1\rangle \Rightarrow a|0\rangle + b|1\rangle = |\psi\rangle$$

Bob manages to get $|\psi\rangle$ from Alice
Without direct connection and only
classical connection "sending correction bits"